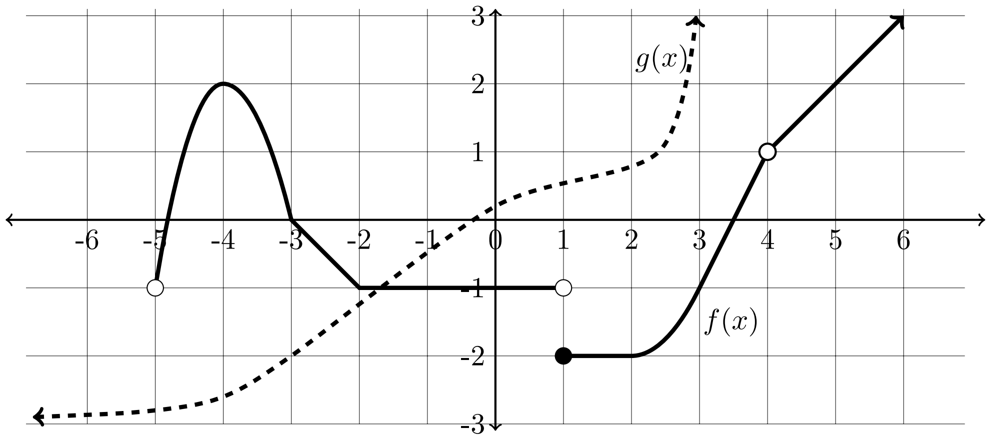
MT 1 Review Exercises

This is a selection of exercises drawn from previous exams for you to practice on.

(1) (10 points) The graphs of two functions $f(x)$, shown thick, and $g(x)$, shown dashed, are given below. Determine the following. If the value does not exist, write "DNE".



(a) $\lim\limits_{x\to -5^+} f(x) =$ ________

(b) $\lim\limits_{x\to -3} f(x) =$ ________

(c) $\lim\limits_{x\to \infty} f(x) =$ ________

(d) $\lim\limits_{x\to 1} f(x) =$ ________

(e) $f'(-4) =$ ________

(f) $\lim\limits_{x\to 3^-} g(x) =$ ________

(g) $\lim\limits_{x\to -\infty} g(x) =$ ________

(h) $\lim\limits_{x\to -3} f(g(x)) =$ ________

(i) What is the domain of $f(x)$? Give your answer in interval form.

(j) Where in the domain of $f(x)$ is the function continuous? Give your answer in interval form.

(2) Evaluate the following limits. Justify your answers with words and/or any relevant algebra. Be sure to use proper notation, as points will be deducted for not doing so.

(a) $\lim\limits_{x\to -5} \dfrac{x^2-25}{x^2+2x-15}$

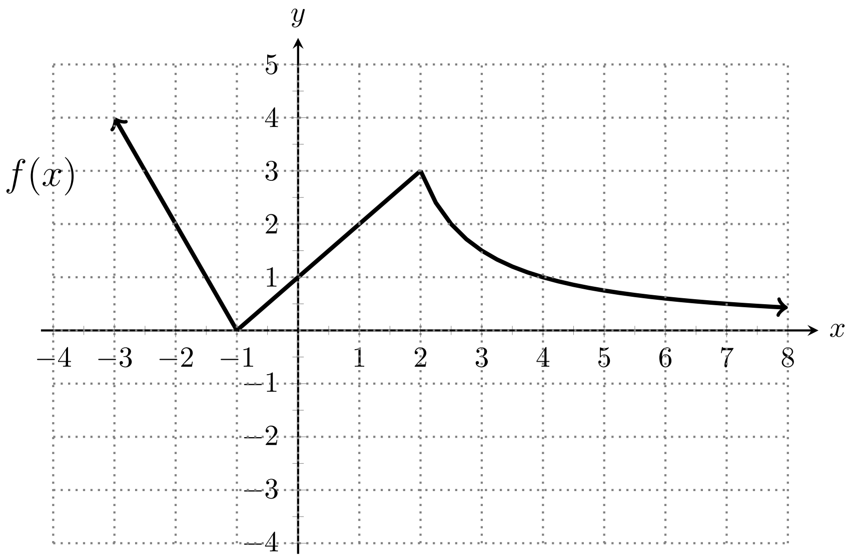
(b) $\lim\limits_{t\to -3} \dfrac{6+4t}{t^2+1}$

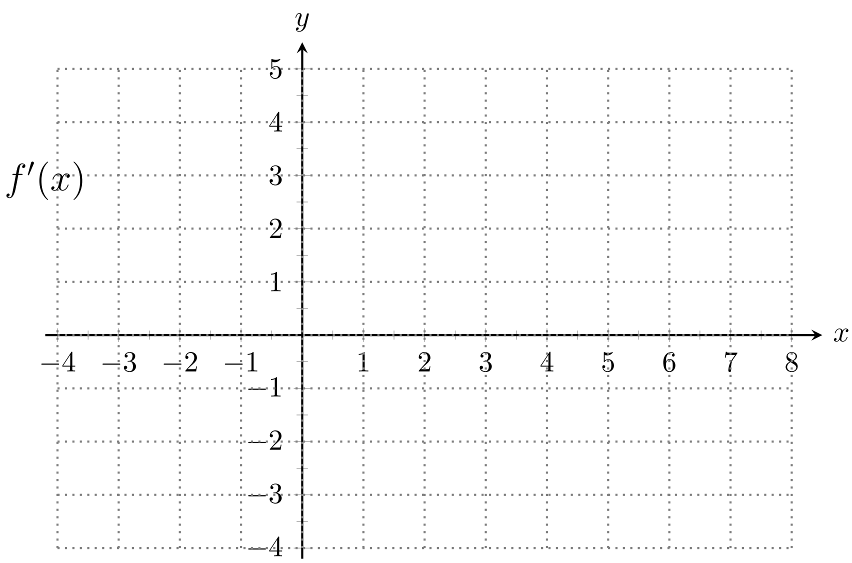
(c) This is a two parter:

(i) $\lim\limits_{x\to -1^-} \sqrt{x^2-1}$

(ii) Why do we not evaluate $\lim\limits_{x\to -1^+} \sqrt{x^2-1}$? Explain using a sentence.

(3) Use the graph of $f(x)$ (on top) to sketch the graph of $f'(x)$ (on bottom).





(4) An icicle is growing at the back of a house. At time $t \geq 0$ days the length of the icicle is

$$\ell(t) = \frac{14t + 4}{t + 2}$$

inches. Answer the following questions about the function $\ell(t)$ and be sure to include **units** in your answers.

(a) Compute the average rate of change of the length of the icicle from time $t = 0$ to time $t = 2$ days. **Do not forget units!**

(b) It is a known fact that $\ell'(t) = \dfrac{24}{(t+2)^2}$. Compute $\ell'(2)$ (**with units**) and explain what this quantity means about the icicle.

(c) Compute $\lim_{t\to\infty} \ell(t)$. Then explain what this quantity means in precise but everyday language that the general public would understand. Do not forget units!

(5) Is the following function continuous at $x = 3$? Justify your answer.

$$f(x) = \begin{cases} \frac{6-2x}{x-3} & \text{if } x \leq 3 \\ 2 & \text{if } x = 3 \\ x - 5 & \text{if } x \geq 3 \end{cases}$$

(6) Let $H(t)$ be the daily cost, in dollars, of heating a building when the outside temperature is t degrees Fahrenheit.

(a) What is the meaning of $H(0) = 50$? Your answer should be a complete sentence and must include units.

(b) What are the units of $H'(t)$?

(c) Do you expect $H'(t)$ to be positive or negative? **Justify** your answer using complete sentences.

(7) Consider the function

$$f(x) = \frac{1}{6-x}.$$

Find $f'(5)$ using the **limit definition of the derivative** and show your work using all appropriate notation.

No credit will be awarded for using other methods. Begin by writing down the limit definition of the derivative.

(8) For each of the following functions, compute the derivative.

You do not need to simplify your answers.

(a) $y = 11x^3 - \frac{4x}{3} + x^2 + x^{5/3}$

(b) $a(\theta) = \theta^3 \cos(\theta)$

(c) $f(t) = t\sqrt{t} - \dfrac{1}{8t^4} + \sqrt{2}$

(9) The questions below reference the function $f(x) = (3 - x)(x - 7)$.

(a) Find the slope of the line tangent to at $x = 4$.

(b) Sketch the tangent line to $f(x)$ at $x = 4$ on the graph of $f(x)$ below.